<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="4">Marks Available</th><th rowspan="2">Maths</th><th rowspan="2">Prac</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th></tr>
<tr><td>2.</td><td>(a)</td><td></td><td></td><td>The vibrations / oscillations (1)<br>are parallel to the direction of wave or travel or energy transfer (1)</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>Speeds of 9 OR 5 [km/s] OR 0.5 × 8 squares (1)<br>Difference = 4 [km/s] (1)</td><td></td><td>2</td><td></td><td>2</td><td>1</td><td></td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td></td><td>[Maximum] speed in the mantle is greater than the speed in the outer core / greatest speed is in the mantle / 15 [km/s] and 13 [km/s] (1)<br>Mantle acts like a solid / outer core is liquid (1) so Bob's claim is true.<br>Conclusion must be included to award 2 marks</td><td></td><td></td><td>2</td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>Mean speed = $\frac{6\,300}{550}$ (1)<br>= 11.45 (km/s) (1)<br>This is the actual speed at depth of 1 200 km (or at 4 900 km) (1)<br>[So Bob's statement is not true.]<br><br>Alternative for third mark:<br>[At 3 500 m] the speed is 10.0 [km/s] [So Bob's statement is not true.]<br><br>Alternative:<br>Speed at 3500 km is 10.0 [km/s] (1)<br>Time = $\frac{6\,300}{10}$ (1)<br>= 630 [s] (1) [So Bob's statement is not true.]<br><br>Alternative:<br>Speed at 3500 km is 10.0 [km/s] (1)<br>Distance = 10 × 550 (1)<br>= 5 500 [km] [So Bob's statement is not true.]</td><td></td><td></td><td>3</td><td>3</td><td>2</td><td></td></tr>
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| Question | | | | Marking details | Marks Available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | (d) | (i) | | 2 × 550 = 1 100 [s] | | 1 | | 1 | 1 | |
| | | (ii) | | P wave only shown i.e. one cycle (1)<br>Size no bigger than wave at B (1)<br>Position, the start of the wave must be within the correct 200 s range based on (d)(i) (expect 1 000 – 1 200) or apply an **ecf** (1)<br>N.B.<br>If drawn correctly but at station A or B apply a 1 mark penalty | | | 3 | 3 | | |
| | | | | **Question 2 total** | **2** | **3** | **8** | **13** | **4** | **0** |